

Online Learning (Adaptive Hedge)

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Abstract

In this paper, I compare the implementations of the Hedge and Adaptive Hedge algorithms in three different settings of synthetic losses. The deliberately diverse natures of synthetic losses provides an opportunity to study the behavior of these algorithms depending on the difficulty of the environment. This was achieved by analyzing the regret versus time plots for Adaptive Hedge algorithm with adaptive learning rate and Hedge algorithm with fixed learning rate. The results showed that AdaHedge adapts well to the difficulty of the problem: in "easy instances" it has smaller regret than Hedge while simultaneously guaranteeing worst case regret bounds.

1 Introduction

This paper is focused on DTOL, Decision Theoretic Online Learning. It is about learning problems where the information is received in rounds. The problem involves a fixed amount of K actions, the task of the agent is following: at first round all actions receive equal probability by the agent, then, all K actions incur some loss in $[0, 1]$, and by multiplying them with weights assigned by the agent we get expected loss of the agent. To evaluate the performance of the agent we analyze its *regret*, that is, the difference between agent's accumulated loss and accumulated loss of best action at that round. Hence, the agent's goal is the minimization of regret to get closest accumulated loss to the best action in hindsight.

There are many algorithms that use different ways on how to assign the weights of the next round, this step uses the accumulated losses of all actions. The Hedge and Adaptive Hedge algorithms calculate the next round weights by using the *learning rate* η . The difference between these two methods is that Hedge algorithm needs to tune this parameter beforehand, whereas Adaptive Hedge does so itself during the learning problem. Since Hedge is so dependent on tuning of η , there may be some drawbacks, such as, a learning rate that is too large can be sensitive to noise and unstable, too small can have slower adapting. η can be tuned if number of rounds T is known, so that it achieves worst case regret bounded by $\sqrt{T \ln(K)/2}$; if best action's cumulative loss L_T^*

is known, regret has bound $\sqrt{2L_T^* \ln(K)} + \ln(K)$, which can be lower if L_T^* is small[1].

This difficulty of tuning the learning rate can be eased by using Adaptive Hedge that tunes the parameter based on the observed difficulty of the problem, and performs well in "easy instances" where there is one consistent best expert and in adversarial settings too.

2 Algorithms

2.1 Hedge Algorithm with fixed learning rate

This paper analyzes Hedge with fixed learning parameter η . It is crucial since it impacts how aggressively Hedge updates its weights:

$$w_{i,t+1} = \frac{w_{i,t} e^{-\eta \ell_{i,t}}}{\sum_{j=1}^K w_{j,t} e^{-\eta \ell_{j,t}}} \quad (1)$$

where $w_{i,t+1}$ is the weight of agent i in next round $t+1$ calculated by using weight and observed loss of agent at current round t . In all experiments I compared regret curves of Hedge with three different tuning ways of η , assuming T, L_T^* were known:

$$\eta_1 = \sqrt{\frac{2 \ln(K)}{T}}; \quad (2)$$

$$\eta_2 = \sqrt{\frac{2 \ln(K)}{L_T^*}}; \quad (3)$$

$$\eta_3 = \ln\left(1 + \sqrt{\frac{2 \ln(K)}{L_T^*}}\right) \quad (4)$$

which were obtained from the paper of van Erven et al. [1].

2.2 Adaptive Hedge Algorithm

Adaptive Hedge uses doubling trick by dividing the rounds into segments, and at the start of each such new segment, learning parameter η is scaled down by $\phi > 1$. In my algorithm i chose initial $\eta = 1$ and $\phi = 2$. It determines how to start a new segment by monitoring when the cumulative mixability gap $\Delta_T(\eta)$ exceeds budget $b = (\frac{1}{e-1} + \frac{1}{\eta}) \ln(K)$.

The mixability gap is key to measuring the difficulty of the problem. It does so by measuring how closely an ideal lower bound for actual loss of any prediction strategy is approached.

The coding of the algorithm in the experiments closely followed the pseudo-code in [1] and was also modified to include the output of the regret curve of the agent:

Algorithm 1 AdaHedge(ϕ)

Require: $\phi > 1$

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1:  $\eta \leftarrow \phi$ 
2: for  $t = 1, 2, \dots$  do
3:   if  $t = 1$  or  $\Delta \geq b$  then
4:     Start a new segment
5:      $\eta \leftarrow \eta/\phi$ 
6:      $b \leftarrow \left(\frac{1}{e-1} + \frac{1}{\eta}\right) \ln(K)$ 
7:      $\Delta \leftarrow 0$ 
8:      $w \leftarrow (w^1, \dots, w^K) \leftarrow (\frac{1}{K}, \dots, \frac{1}{K})$ 
9:   end if
10:  Make a decision
11:  Output probabilities  $w$  for round  $t$ 
12:  Actions receive losses  $\ell_t$ 
13:  Prepare for the next round
14:   $\Delta \leftarrow \Delta + w \cdot \ell_t + \frac{1}{\eta} \ln(w \cdot e^{-\eta \ell_t})$ 
15:   $w \leftarrow \frac{(w^1 e^{-\eta \ell_t^1}, \dots, w^K e^{-\eta \ell_t^K})}{w \cdot e^{-\eta \ell_t}}$ 
16: end for
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3 Experiments

Stochastic Synthetic Losses

Stochastic synthetic losses were generated for independent experts $K = 4$. All four actions were assigned Bernoulli losses; however, one of the experts had parameter 0.3, while others had Bernoulli parameter equal to 0.5. Thus, one expert had much lower cumulative loss compared to other three, creating an example of an "Easy Instance" where the best expert had a large advantage. In Figure 1(a), we can clearly observe how the cumulative loss of the best action linearly separates from others showing a clear divergence. The number of rounds was set to $T = 10000$ and I did 10 runs/repetitions of the experiment to smooth out the regret curve.

Synthetic Losses with Low Gap

Another case suggested for the project is the low gap between expectations of the normal distributions of actions. The experiment had $K = 4$ independent actions of normal distribution, where mean value of the first action was equal to 0.49, while other three actions had μ equal to 0.5. I chose standard deviation to be $\sigma = 0.05$ for all four actions. $T = 10000$ rounds and 10 repetitions were conducted. Action one has slightly lower cumulative loss in Figure 1(b).

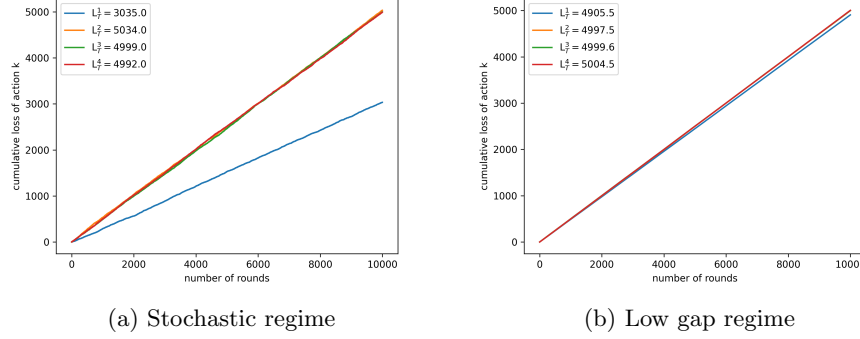


Figure 1: Synthetic losses

Adversarial Synthetic Losses

In the adversarial regime, losses were generated using Bernoulli losses for $K = 3$ actions in 10000 rounds, but this time the best experts were different in different blocks:

- In rounds 1–5000, best action 1 had Bernoulli parameter 0.4, while actions 2 and 3 had parameter 0.5.
- In rounds 5000–10000, best action 2 had Bernoulli(0.2) losses, and actions 1 and 3 both had Bernoulli(0.5) losses.

This experiment aimed at analyzing the performances of AdaHedge and Hedge in a setting where best action was not stationary and how this switch affects their regret curves. Cumulative loss plot in Figure 2:

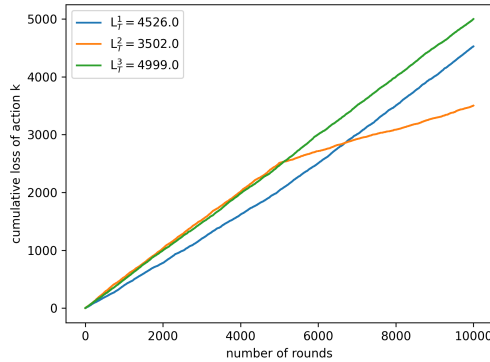


Figure 2: Adversarial regime

4 Discussion and Results

4.1 Stochastic Synthetic Losses

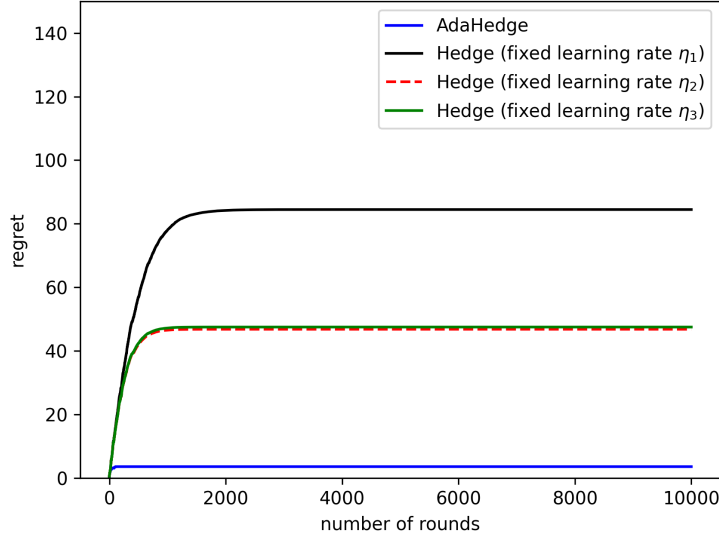


Figure 3: Regret curve for stochastic regime

This experiment constitutes as an "easy instance" where one action consistently has lower expected loss. As Figure 3 clearly demonstrates, in this experiment, AdaHedge algorithm achieved lowest average regret curve, since it is able to quickly concentrate probability mass on best expert and regret growth slows faster. All algorithms appear to incur bounded regret over tested horizon, however, for Hedge algorithms, the learning rate is too small and reacts too slowly. If we were to compare the multiple η choices for the Hedge, lowest regret curve corresponds to $\eta_2 = \sqrt{\frac{2\ln(K)}{L_T^*}}$, then slightly worse $\eta_3 = \ln(1 + \sqrt{\frac{2\ln(K)}{L_T^*}})$ and finally $\eta_1 = \sqrt{\frac{2\ln(K)}{T}}$ incurs almost twice larger regret bound than η_2 and η_3 , because it depends on large horizon. When L_T^* is small, η_2 and η_3 adapt better. The same pattern is observed in all experiments with η_2 having lowest regret curve.

4.2 Synthetic Losses with Low Gap

This part of the experiment again showcases the strength of AdaHedge algorithm that is able to quickly identify the best expert in a setting where the best

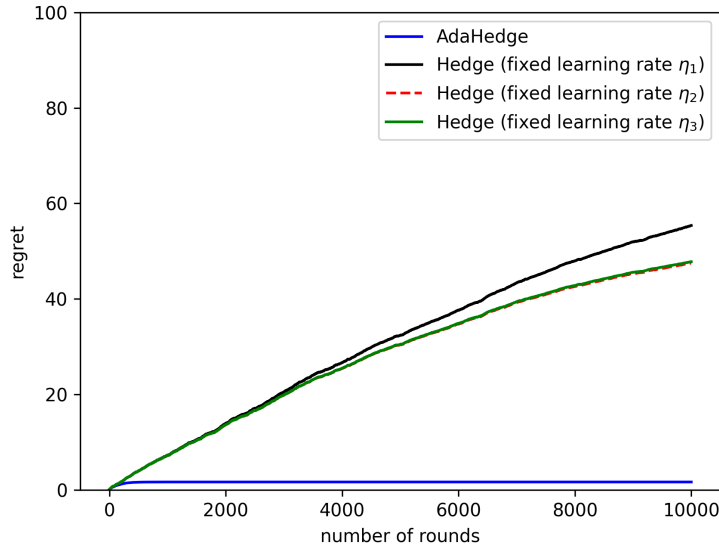


Figure 4: Regret curve for low gap regime

expert has a lower cumulative loss gap with other actions. The regret curve of AdaHedge is bounded on the tested horizon.

Interestingly, regret curves of the Hedge algorithm for all three learning rates approach the regret bound much slower than in the previous experiment, because the differences in accumulated loss of experts is smaller and the signal is weaker. Thus, weight updating cannot concentrate quickly on best expert compared to previous regime and regret accumulation and approaching of theoretical bound happens at a slower rate.

4.3 Adversarial Synthetic Losses

The last setting is when best expert changed over time. Here too, Adaptive Hedge repeatedly obtained significantly lower regret than Hedge implementations for all three η -s (Figure 5). The online adaptation demonstrated better performance without tuning the parameter.

Hedge with fixed η is effective only if it is well tuned, which is the issue AdaHedge improves on. Especially, in such adversarial cases where best expert switches or as in 4.2, in environments where loss gaps are small between experts.

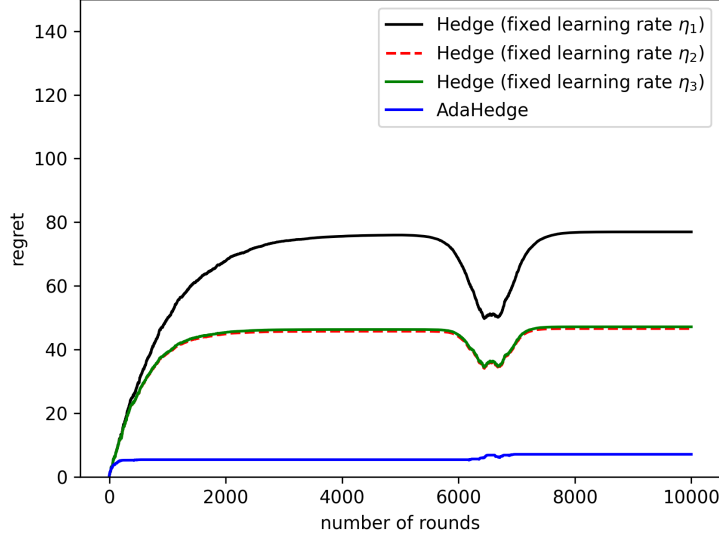


Figure 5: Regret curve for adversarial regime

5 Conclusion

The results of all three experiments with synthetic losses confirm the theoretical strength of Adaptive Hedge, the dynamical adjusting of learning parameter η is a better tool for Online Learning than fixed learning rate. Instead of relying on a manually selected η AdaHedge improves by automating the selection process according to observed losses, and obtains lower cumulative regret compared to Hedge with fixed learning rates tuned as in the aforementioned formulas (2)–(4), which require information on T or L_T^* .

The main advantage of Adaptive Hedge is that it has better performance in "easy instances" and still guarantees worst case regret bounds.

References

- [1] van Erven, T., Grünwald, P., Koolen, W. M., and de Rooij, S. (2011). Adaptive Hedge. arXiv. <https://doi.org/10.48550/ARXIV.1110.6416>